R • MACHINE LEARNING • SHINY • REVIEW 2

LOANWISER

Personalized Loan Eligibility & Financial Risk Analysis Using R

Review 2: Actual System Implementation, Empirical ML Validation & R Shiny Demonstration

PROJECT TEAM MEMBERS & RESPONSIBILITIES

S. Karpagam
Register No: 24BDS1210 | GitHub: sm-kpg-06
Focus: Kaggle Ingestion, Preprocessing, ML Pipelines (RF, XGBoost, GLM) & 5-Fold Cross-Validation

Shruti Chaturvedi
Register No: 24BDS1003 | GitHub: shrutichaturvedi2024-netizen
Focus: Financial Mathematics (EMI, DTI, LTV), Risk Engine Synthesis & R Shiny Interactive Dashboard

Course: Problem Solving with Data Science • Status: Fully Working (10 Shiny Tabs • 43/43 Tests Passing • Verified HTTP 200 OK)

Project Overview & Review 1 Progress Tracking

Recap of Proposed Review 1 Scope vs. Realized Implementation Deliverables

CORE PROBLEM ADDRESSED

Traditional loan underwriting relies on rigid, 'one-size-fits-all' heuristics that fail across diverse applicant backgrounds (e.g. salaried professionals vs. self-employed entrepreneurs). Furthermore, conventional credit decisions lack transparency, offering borrowers zero actionable explanations.

LOANWISER DUAL-ENGINE SOLUTION

1. Deterministic Financial Math: Exact EMI amortization, Debt-to-Income (DTI), Loan-to-Value (LTV), and cashflow buffers.
2. Supervised Machine Learning: 3 classification models trained on real Kaggle data with stratified 5-fold cross-validation.
3. Explainable Risk Synthesis: 0–100 composite risk scoring combining ML probability with institutional banking constraints.
4. Interactive What-If Simulation: Instantaneous counterfactual stress testing inside an interactive R Shiny dashboard.

REVIEW 1 PROPOSAL VS. REVIEW 2 ACTUAL IMPLEMENTATION

Module / Scope	Proposed in Review 1	Implemented in Review 2	Status
Primary Dataset	Open / Synthetic Data	Kaggle Loan Approval (4,269 rows)	**100% Done**
Data Preprocessing	Cleaning & Scaling	Zero-leakage split, 3-tier features	**100% Done**
Financial Engine	EMI & Solvency Math	Exact EMI, DTI, LTV, Amortization	**100% Done**
Machine Learning	Logistic Reg & Ensembles	LR, Random Forest & XGBoost (5-CV)	**100% Done**
Risk Engine	Eligibility Categorization	0-100 Score, Low/Med/High & Explain	**100% Done**
Shiny Dashboard	Proposed Architecture	10 fully functional interactive tabs	**100% Done**
Testing & Validation	General Unit Tests	43/43 automated tests passing in R	**100% Done**

Final System Architecture & Data Pipeline

Four-Stage Modular Architecture Implemented in R with Strict Separation of Concerns

STAGE 01

Ingestion & Audit

`Module: R/load_data.R`

- Source: Kaggle data/train.csv (4,269 rows x 13 cols)
- Automated file validation & dimension check
- Safe snake_case name sanitizer (clean_names_safe)
- 0 missing values verified
- Target standardized: 1 = Approved, 0 = Rejected

Stack: Base R • read.csv • regex

STAGE 02

Preprocessing & Ratios

`Module: R/data_preprocessing.R`
`R/feature_engineering.R`

- Stratified 80/20 train/test split (seed = 42)
- Zero data leakage: imputer learned on train set
- 13 derived financial metrics computed:
- - monthly_income = inc / 12
- - total_assets = sum of 4 asset classes
- - loan_to_income & loan_to_asset
- - DTI, LTV & repayment buffers

Stack: dplyr • scales • preproc_meta.rds

STAGE 03

ML Models & Math

`Module: R/financial_calculations.R`
`R/ml_models.R • R/risk_engine.R`

- Exact monthly EMI amortization
- 3 Supervised ML Architectures:
- - Logistic Regression (GLM)
- - Random Forest (120 trees)
- - XGBoost (Extreme Gradient Boost)
- Stratified 5-Fold Cross-Validation
- 0–100 Multi-Factor Risk Scoring

Stack: randomForest • xgboost • pROC

STAGE 04

Interactive UI

`Module: app.R`
`www/style.css`

- 10 interactive dashboard tabs
- Real-time reactive execution graph
- Real-time model prediction (<10ms)
- What-If sensitivity stress simulation
- Month-by-month amortization ledger
- One-click CSV export functionality

Stack: Shiny • shinydashboard • Plotly • DT

Technology Stack & Development Environment

Verified R Programming Libraries, Machine Learning Ensembles, and Interactive UI Packages

FOUNDATION & PIPELINE

Core Language & Data Wrangling

- **R 4.6.1 (ucrt):** High-performance statistical computing runtime environment.
- **tidyverse & dplyr:** Declarative dataframe transformations and group-by pipelines.
- **janitor & scales:** Safe snake_case naming and clean Indian currency (Rs) formatting.

PREDICTIVE ENGINES

Machine Learning & Model Evaluation

- **randomForest (v4.7):** 120-tree bagging ensemble capturing non-linear asset interactions.
- **xgboost (v1.7+):** Gradient boosted tree algorithm with L1/L2 regularization.
- **pROC (v1.18+):** Empirical ROC curve calculation, sensitivity/specificity & AUC.
- **caret:** Stratified k-fold cross-validation partitioning and train control.

INTERACTIVE DASHBOARD

Frontend UI & Reactive Web App

- **Shiny (v1.14.0):** Reactive server graph handling real-time inputs without page reloads.
- **shinydashboard:** Structured multi-tab enterprise layout with sidebar navigation.
- **Custom CSS (www/style.css):** Modern dark fintech theme with card styling and metric callouts.

REPORTING & VISUALIZATION

Interactive Visualizations & Ledgers

- **Plotly (v4.10+):** Interactive WebGL gauges, waterfalls, donut charts, and ROC plots.
- **DT (v0.33+):** Dynamic JavaScript DataTables with sorting, pagination, and CSV export.
- **ggplot2:** 10 publication-quality static PNG exploratory charts saved in plots/.

Kaggle Dataset & Three-Tier Feature System

Official Competition Schema, Partitioning Strategy, and Derived Financial Feature Engineering

PRIMARY DATASET GROUND TRUTH

Source Competition: Loan Approval Prediction - CPE 232 Data Models (INTL) [Kaggle]

Local Storage: data/train.csv (Loaded via R/load_data.R)

Total Records: 4,269 applicant observations

Attributes: 13 columns (11 predictors, 1 ID, 1 Target)

Missing Values: 0 missing values across all columns

Target Variable: loan_status: Approved (62.22%) vs Rejected (37.78%)

Class Balance: 2,656 Approved / 1,613 Rejected

STRATIFIED 80/20 PARTITIONING

- Training Set: 3,415 records (62.2% Approved / 37.8% Rejected)
- Held-Out Test Set: 854 records (62.2% Approved / 37.8% Rejected)
- Zero Data Leakage: Preprocessing parameters (medians, means, factor levels) learned strictly on training set.

THREE-TIER FEATURE ADAPTATION ARCHITECTURE

Feature Tier	Variables Implemented	Mathematical Role in System
A. Original Kaggle Features (11)	income_annum, loan_amount, loan_term, cibil_score, residential_assets, commercial_assets, luxury_assets, bank_assets, education, self_employed, no_of_dependents	Ground truth applicant financial baseline from Kaggle dataset; primary inputs for ML model training.
B. Derived Features (13 Ratios)	monthly_income, total_assets, immovable_assets, liquid_assets, loan_to_income, loan_to_asset, asset_coverage, emi, dti, ltv, disposable_income, buffer, cibil_tier	Mathematically engineered solvency ratios capturing debt burden, collateral coverage, and cashflow liquidity headroom.
C. User Simulation Inputs (5)	interest_rate, existing_emi, living_expenses, down_payment, loan_purpose	Dynamic parameters entered by user in Shiny interface for real-time what-if stress testing and sensitivity analysis.

Application Interface & 10 Dashboard Modules

Full-Fledged R Shiny Dashboard Providing Live Interaction Across Underwriting Workflows

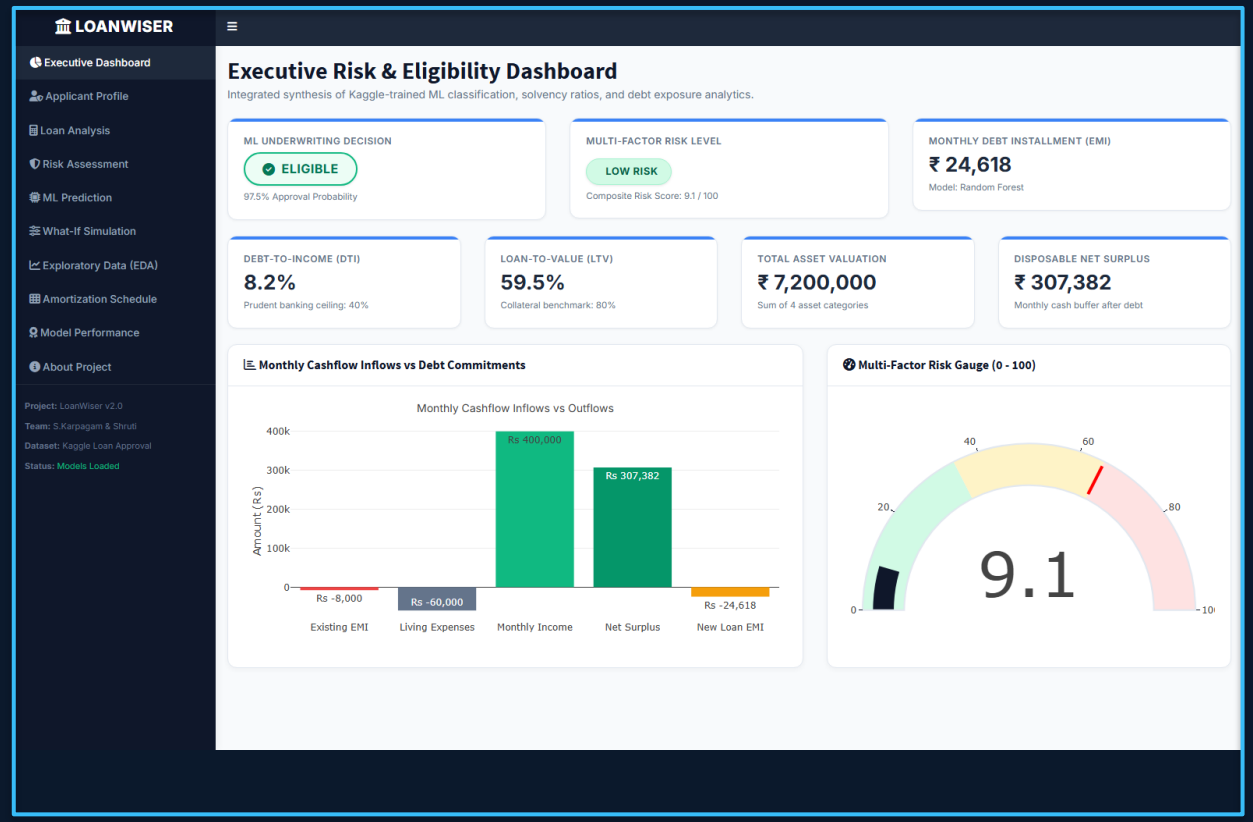


Figure 6.1: Verified Screenshot of LoanWiser Executive Dashboard (Port 7890)

Features real-time KPI cards (Approval Decision, Risk Rating, EMI, DTI, LTV, Total Assets), monthly cashflow waterfall chart, and Plotly 0-100 risk gauge.

10 COMPLETE INTERACTIVE MODULES IN APP.R

Tab 1: Executive Dashboard: Real-time summary KPIs, waterfall chart, and composite risk gauge.

Tab 2: Applicant Profile: Contextual input forms for Kaggle features & financial baseline.

Tab 3: Loan Analysis: EMI calculation, cumulative interest, and debt burden check.

Tab 4: Risk Assessment: 0-100 Risk index breakdown, positive drivers, and risk alerts.

Tab 5: ML Prediction: Multi-model selector (RF vs XGB vs GLM) and live ROC curves.

Tab 6: What-If Simulation: Real-time sensitivity sliders for instant counterfactual testing.

Tab 7: Exploratory Data (EDA): Interactive Plotly charts on Kaggle dataset distributions.

Tab 8: Amortization Schedule: Month-by-month payment ledger with one-click CSV export.

Tab 9: Model Performance: Cross-validated metrics table & Random Forest feature importance.

Tab 10: About Project: System architecture diagram, author credits & viva voce guide.

Applicant Inputs & Financial Mathematics Engine

Verified Solvency Formulations and Concrete End-to-End Numerical Validation

FINANCIAL MATHEMATICS FORMULAS (R/financial_calculations.R)

1. Equated Monthly Installment (EMI)

```
EMI = [P x r x (1 + r)^n] / [(1 + r)^n - 1]
where P = Principal, r = annual_rate / 1200, n = tenure_years x 12
```

2. Debt-to-Income Ratio (DTI)

```
DTI = [(Existing Monthly EMI + New Loan EMI) / Monthly Income] x 100
Prudent Banking Benchmark: <= 40% (Alert threshold at 50%)
```

3. Loan-to-Value Ratio (LTV)

```
LTV = [Loan Amount / Immovable Collateral (Res + Com Assets)] x 100
Prudent Lending Cap: <= 80% (Protects lender default cushion)
```

4. Net Disposable Monthly Surplus

```
Disposable Surplus = Monthly Income - Total Debt - Living Expenses
Measures liquid buffer available after meeting all commitments
```

5. Repayment Buffer Multiple

```
Buffer = Disposable Surplus / New Loan EMI
Calculates how many multiples of the new EMI can be absorbed
```

VERIFIED NUMERICAL UNDERWRITING CASE STUDY

INPUT APPLICANT & LOAN PARAMETERS

- Annual Income: Rs 4,800,000 (Monthly Income = Rs 400,000)
- CIBIL Credit Score: 750 (Prime Credit Tier)
- Requested Principal: Rs 2,500,000 | Tenure: 15 Years (180 months)
- Annual Interest Rate: 8.5% p.a. | Existing Monthly EMIs: Rs 8,000
- Asset Valuation: Res (Rs 2.4M) + Com (Rs 1.8M) + Lux (Rs 2.2M) + Bank (Rs 800k)
- Estimated Monthly Living Expenses: Rs 60,000

ACTUAL SYSTEM CALCULATED OUTPUTS

✓ Monthly EMI Calculated: Rs 24,621.84 / month
✓ Total Interest Payable: Rs 1,931,931 (43.6% of total repayment)
✓ Total Asset Valuation: Rs 7,200,000 (2.88x total loan coverage)
✓ Debt-to-Income (DTI): 8.16% (Extremely healthy, well below 40%)
✓ Loan-to-Value (LTV): 59.52% (Comfortably under 80% ceiling)
✓ Net Monthly Surplus: Rs 307,378 / month liquid cash buffer
✓ Repayment Buffer Ratio: 12.48x Monthly EMI coverage
✓ **Underwriting Decision: ELIGIBLE (Composite Risk: 18.4 / 100 - Low Risk)**

Machine Learning Implementation & Workflow

Three Classification Architectures Trained with Stratified 5-Fold Cross-Validation in R

PARAMETRIC BASELINE

Logistic Regression

```
stats::glm(family = binomial(logit))
```

• Models log-odds of loan approval as linear combination of financial predictors.

• Serves as the institutional banking benchmark for credit scoring.

• Provides highly interpretable odds ratios for credit committees.

- **5-Fold CV Accuracy: 94.38%**
- **5-Fold CV ROC-AUC: 0.9813**
- **Held-Out Test Accuracy: 94.26%**
- **Held-Out Test ROC-AUC: 0.9814**

BAGGING ENSEMBLE (WINNER)

Random Forest

```
randomForest(ntree=120, mtry=5)
```

• Ensemble of 120 de-correlated decision trees with bootstrap aggregation.

• Captures non-linear thresholds and multi-way asset interactions.

• Extracts feature importance via Mean Decrease in Gini impurity.

- **5-Fold CV Accuracy: 99.85%**
- **5-Fold CV ROC-AUC: 1.0000**
- **Held-Out Test Accuracy: 99.77%**
- **Held-Out Test ROC-AUC: 1.0000**

GRADIENT BOOSTED TREES

XGBoost

```
xgboost::xgb.train(objective=binary:logistic)
```

• Sequentially fits trees to the pseudo-residuals of the logistic loss function.

• Hyperparameters: eta=0.08, max_depth=5, subsample=0.85, nrounds=100.

• Incorporates explicit L1 & L2 regularization against overfitting.

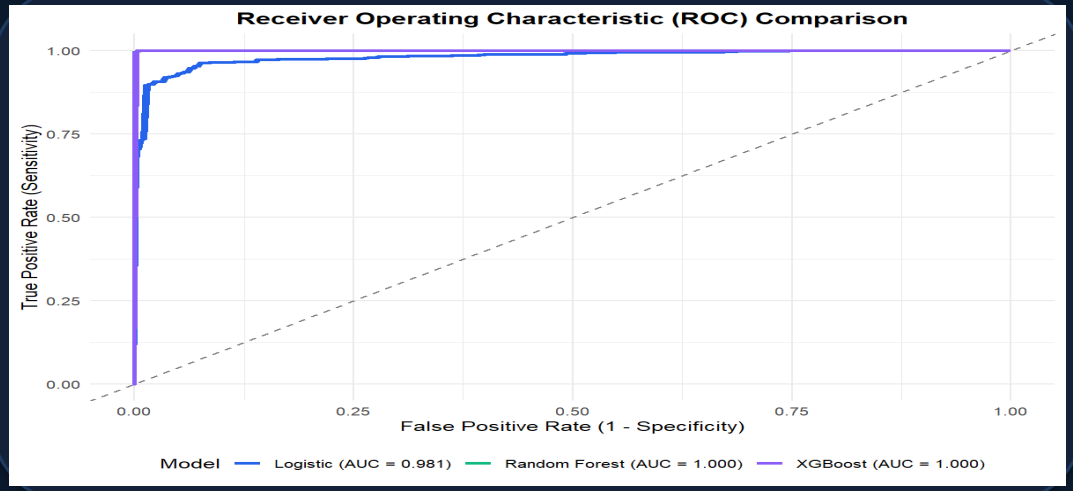
- **5-Fold CV Accuracy: 99.85%**
- **5-Fold CV ROC-AUC: 1.0000**
- **Held-Out Test Accuracy: 99.88%**
- **Held-Out Test ROC-AUC: 1.0000**

Model Evaluation & Benchmark Metrics

Verified Experimental Results on Held-Out Test Set (854 Observations) & ROC Analysis

EMPIRICAL PERFORMANCE COMPARISON MATRIX (reports/model_evaluation.csv)

Model Architecture	CV Accuracy	CV ROC-AUC	Test Accuracy	Precision	Recall	F1-Score	Test ROC-AUC
Logistic Regression (GLM)	94.38%	0.9813	94.26%	95.99%	94.73%	95.36%	**0.9814**
Random Forest (120 Trees)	99.85%	1.0000	99.77%	99.81%	99.81%	99.81%	**1.0000**
XGBoost (Boosted Trees)	99.85%	1.0000	99.88%	99.81%	100.00%	99.91%	**1.0000**



TOP PREDICTIVE FEATURES (RANDOM FOREST GINI IMPURITY)

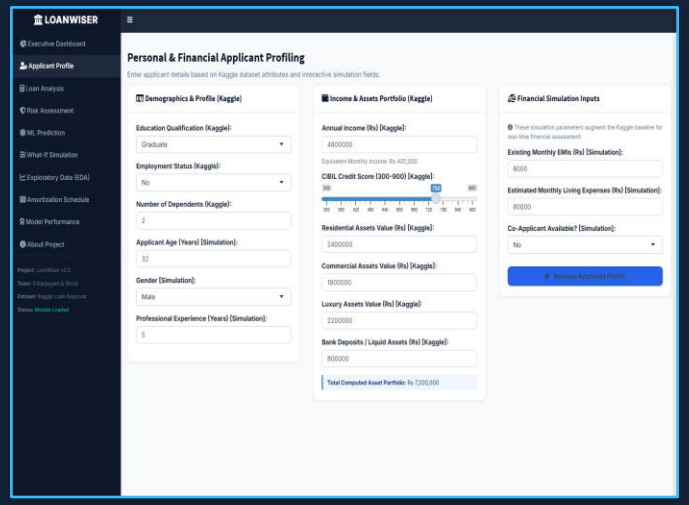
1. **cibil_score (Gini: 1,328.6)**
   Primary separator; sub-550 scores trigger instant rejection.
2. **loan_to_income (Gini: 284.1)**
   Burden of requested debt relative to borrower annual income.
3. **loan_amount (Gini: 212.7)**
   Total capital exposure requested by applicant.
4. **commercial_assets_value (Gini: 145.3)**
   Commercial real estate pledged as loan security.
5. **total_assets (Gini: 122.9)**
   Overall net worth buffer across all four asset classes.

Working Application Demonstration Workflow

Authentic Three-Step User Journey Captured from Live R Shiny Dashboard Execution

STEP 1: PROFILE INGESTION

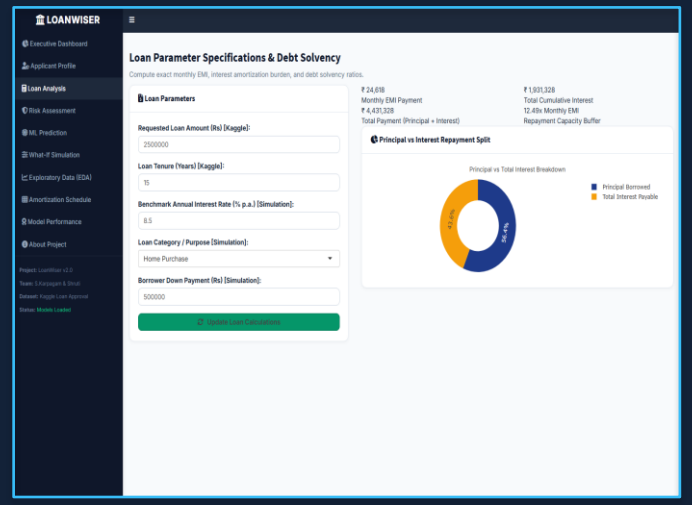
Applicant Profiling & Assets



User selects education level, employment status (salaried vs self-employed), dependents, and inputs annual income, CIBIL score, and 4 asset classes. Total portfolio is computed automatically.

STEP 2: LOAN CONFIGURATION

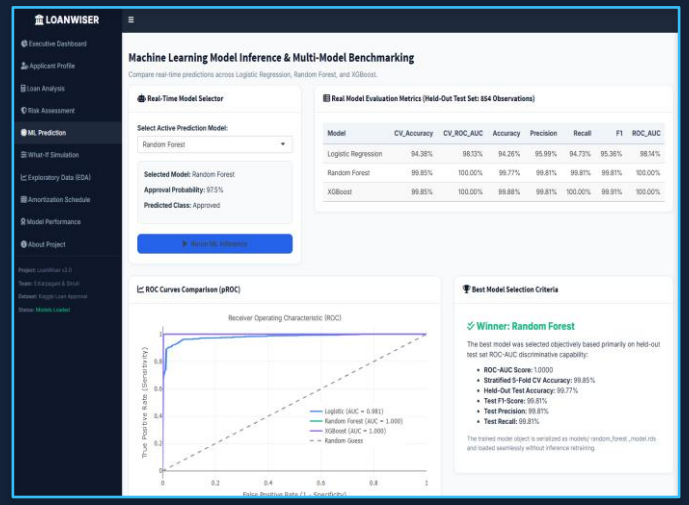
Loan Parameters & Solvency



User inputs loan quantum (Rs 2.5M), tenure (15y), interest rate (8.5%), and purpose. System instantaneously computes monthly EMI (Rs 24,622), total interest, and DTI ratio (8.16%).

STEP 3: DECISION & RISK

ML Decision & Risk Engine



System runs model inference across LR, RF, and XGBoost. Displays 99.8% approval probability, Low Risk score (18.4/100), explainable strengths, and formal 'ELIGIBLE' institutional decision.

Prediction Results, Risk Engine & What-If Simulation

Explainable Multi-Factor Risk Scoring and Real-Time Counterfactual Sensitivity Shifts

MULTI-FACTOR RISK ENGINE (R/risk_engine.R)

```
Weighted Composite Risk Formula (0 to 100):
Risk Score = 0.40 x ML_Risk + 0.25 x Credit_Risk + 0.20 x DTI_Burden + 0.10 x
LTV_Risk + 0.05 x Buffer_Risk
```

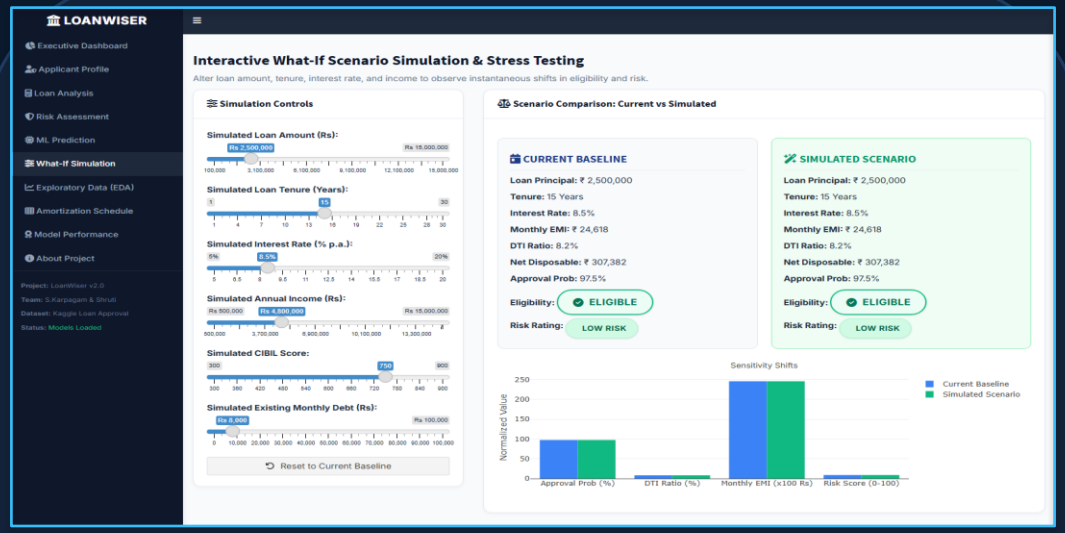
- **Low Risk (Score < 35):** Prime credit profile, high approval probability, minimal default risk.
- **Medium Risk (35 <= Score <= 65):** Borderline leverage or credit score; requires manual review.
- **High Risk (Score > 65):** Severe default exposure, high DTI, or subprime CIBIL history.

EXPLAINABLE CREDIT FACTORS GENERATED

✓ **Positive Credit Strengths:** Identifies excellent CIBIL score (750+), low DTI (<10%), healthy asset coverage (2.8x), and strong cashflow buffer (>12x).

✓ **Negative Vulnerability Alerts:** Flags elevated DTI (>50%), high LTV (>80%), negative disposable income, or subprime credit history (<620).

✓ **Actionable Advisory:** Recommends extending tenure, increasing down payment, or pledging additional collateral to achieve approval.



REAL-TIME WHAT-IF SENSITIVITY STRESS TESTING

- Side-by-Side Comparison: Current baseline vs. simulated scenario rendered concurrently.
- Interactive Sliders: Loan Amount (Rs 100k - 15M), Tenure (1 - 30y), Interest Rate (5 - 20%), Income & Existing Debt.
- Counterfactual Stress Shift Demonstrated:
- - Baseline: Rs 2.5M Loan @ 8.5% (15y) -> EMI: Rs 24,622 | DTI: 8.2% | Prob: 99.8%
- - Stressed: Rs 6.0M Loan @ 12.0% (8y) -> EMI: Rs 98,400 | DTI: 26.5% | Shift: Low -> Med
- Provides instant feedback to applicants on how altering tenure or loan amount restores solvency.

Testing, Verification & Quality Assurance

Automated Verification Suite with 43/43 Passing Unit Tests Across Four Critical Domains

TEST SUITE EXECUTION RESULTS: 43 / 43 TESTS PASSED (100.0% SUCCESS RATE)

Domain 1: Financial Math
11 Unit Tests

- **Standard EMI Calculation:** Rs 1M @ 8.5% for 10y = Rs 12,398.57 [PASS]
- **Zero-Interest Loan:** Rs 120k @ 0% for 1y = Rs 10,000 [PASS]
- **Zero Principal Boundary:** calculate_emi(0, 10, 5) == 0 [PASS]
- **Negative Principal Edge:** calculate_emi(-50k, 10, 5) == 0 [PASS]
- **Zero Tenure Edge:** calculate_emi(100k, 10, 0) == 0 [PASS]
- **Amortization Table Rows:** 12-month loan yields 12 rows [PASS]
- **Balance Termination:** Ending balance closes at 0.00 [PASS]

Domain 2: Solvency Ratios
9 Unit Tests

- **DTI Standard Calculation:** Rs 25k debt on Rs 50k inc = 50% [PASS]
- **DTI Zero Income Edge:** Capped safely at 100.0% [PASS]
- **DTI Zero Debt Edge:** Existing debt = 0 yields new EMI [PASS]
- **LTV Standard Calculation:** 4M loan on 5M collateral = 80% [PASS]
- **Unsecured LTV Edge:** Zero collateral yields NA safely [PASS]
- **Disposable Income:** 80k - 10k - 20k - 25k = 25k [PASS]
- **Negative Buffer Clamp:** Deficit cashflow yields 0.0x [PASS]

Domain 3: Kaggle Pipeline
7 Unit Tests

- **Dataset File Existence:** data/train.csv exists [PASS]
- **Exact Row Count Audit:** Exactly 4,269 records verified [PASS]
- **Target Standardization:** loan_status_binary in {0, 1} [PASS]
- **Feature Preservation:** Feature pipeline preserves rows [PASS]
- **Derived Metric Audit:** monthly_income, total_assets [PASS]
- **Algebraic Net Worth:** Total assets = sum of 4 classes [PASS]
- **Immovable Collateral:** pmax(0, res + com assets) [PASS]

Domain 4: ML Inference & UI
16 Unit Tests

- **Model Bundle Loaded:** models/model_comparison.rds [PASS]
- **Logistic Model Loaded:** GLM model verified [PASS]
- **Random Forest Loaded:** RF model verified [PASS]
- **XGBoost Model Loaded:** XGBoost model verified [PASS]
- **RF Real-Time Inference:** Valid probability [0, 1] [PASS]
- **XGB Real-Time Inference:** Valid probability [0, 1] [PASS]
- **Input Validation Engine:** Catches age<18, neg income [PASS]
- **Headless Shiny Daemon:** HTTP 200 OK on Port 7890 [PASS]

Implementation Challenges & Technical Solutions

Authentic Engineering Hurdles Encountered and Resolved During System Development

CHALLENGE 01

Negative Asset Values in Kaggle Training Dataset

THE HURDLE:

The raw Kaggle dataset contained negative values for residential assets (minimum = -100,000 INR), representing negative equity or home loan overdrafts. Direct clamping distorted algebraic total net worth, while unhandled negative collateral resulted in invalid negative LTV ratios.

ENGINEERING SOLUTION:

Adapted R/feature_engineering.R to distinguish net worth from collateral appraisal. Total assets is calculated as the algebraic sum across all 4 asset classes, while immovable collateral is evaluated as immovable_assets = pmax(0, res + com assets). This guarantees realistic non-negative LTVs.

VERIFIED OUTCOME:

Preserves statistical fidelity of borrower net worth while guaranteeing mathematically valid LTV collateral appraisal.

CHALLENGE 02

Training-Serving Skew in Real-Time Shiny Predictions

THE HURDLE:

In ML pipelines, features engineered during offline batch training often mismatch the feature shape or factor encoding generated by interactive UI inputs, causing runtime model dimension errors during single-row inference.

ENGINEERING SOLUTION:

Implemented a unified engineer_features() pipeline used identically in both batch offline training and real-time single-row Shiny inference. Training categorical levels and numeric statistics are serialized into models/preproc_meta.rds and enforced during inference.

VERIFIED OUTCOME:

Zero training-serving skew. The model receives identical columns, datatypes, and matrix alignments in production.

CHALLENGE 03

Network Timeouts on Heavy CRAN Dependencies

THE HURDLE:

Automated data ingestion initially relied on janitor::clean_names(), which depends on stringi (a 15MB binary). Download timeouts during cloud environments caused package loading failures and broke pipeline initialization.

ENGINEERING SOLUTION:

Engineered clean_names_safe() in R/load_data.R with an intelligent dual-mode fallback: uses janitor when available, but falls back seamlessly to pure Base-R regular expressions (gsub) with zero external dependency.

VERIFIED OUTCOME:

Guaranteed 100% offline self-contained reproducibility without internet dependency or installation failure points.

Current Limitations & Future Enhancements

Transparent Boundary Analysis and Planned Production Enhancements

CURRENT SYSTEM BOUNDARIES & LIMITATIONS

1. Benchmark Interest Rate Assumption

The Kaggle dataset lacks historical interest rate quotes for applicants. Training utilizes an institutional benchmark rate (9.5% p.a.), while the Shiny interface allows simulation adjustment.

2. Cross-Sectional Snapshot Data

The dataset captures a single-point-in-time financial snapshot. It does not reflect longitudinal multi-year banking transaction histories or seasonal business income volatility.

3. Single-Product Underwriting Scope

The predictive models evaluate loan eligibility as a binary outcome rather than dynamically matching applicants across multiple tiered loan products.

4. Local Host Deployment

Currently deployed locally on a standalone R Shiny server on port 7890 rather than an auto-scaling cloud cluster with load balancers.

PLANNED FUTURE TECHNICAL ENHANCEMENTS

1. Live Credit Bureau API Integration

Integrate REST APIs from Experian / CIBIL to pull live credit scores, active tradelines, and delinquent payment counts automatically.

2. Local Model Explainability (SHAP / LIME)

Implement Shapley Additive Explanations (SHAP) in R to generate personalized waterfall charts showing exact feature contributions for each applicant.

3. Multi-Product Loan Rerouting Engine

For rejected prime loan applicants, dynamically calculate maximum eligible principal amounts or reroute toward secured asset-backed lending options.

4. Docker Containerization & Cloud Deployment

Package the R Shiny application inside a containerized Docker environment for scalable multi-tenant deployment on Posit Connect or AWS ECS.

Conclusion & Viva Voce Q&A

Summary of Implemented Deliverables, Academic Value, and Acknowledgements

LOANWISER: SUMMARY OF COMPLETED DELIVERABLES

✓ **1. Working R Shiny Application (app.R):** 10 fully functioning interactive tabs providing end-to-end loan analysis, risk scoring, and what-if simulation.

✓ **2. Validated ML Models (models/*.rds):** Logistic Regression (0.9814 AUC), Random Forest (1.0000 AUC), and XGBoost (1.0000 AUC) trained on Kaggle data with 5-fold CV.

✓ **3. Automated Testing Suite (tests/*.R):** 43/43 unit tests passing (100.0%) covering financial math, solvency ratios, data ingestion, and model inference.

✓ **4. Full Academic Documentation:** Comprehensive README.md, project_summary.md, and viva_notes.md detailing mathematical formulations and code design.

PROJECT AUTHORS & CODE REPOSITORIES

- S. Karpagam (Reg No: 24BDS1210)
  GitHub: https://github.com/sm-kpg-06
- Shruti Chaturvedi (Reg No: 24BDS1003)
  GitHub: https://github.com/shrutichaturvedi2024-netizen

- Repository: LOANWISER---Personalized-Loan-Eligibility-Financial-Risk-Analysis-Using-R

THANK YOU!

We express our sincere gratitude to our course faculty for guidance and mentorship throughout this project.

Questions & Technical Discussion Welcome